

A Crowdsourced-First Calibration Framework for Multi-Class MATSim Models in Data-Constrained Heterogeneous Traffic

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1. Introduction

Agent-based transport simulation is increasingly used to evaluate interventions in rapidly urbanising cities, yet calibration remains difficult where field observations are sparse, and traffic is highly heterogeneous. Indian urban traffic combines motorcycles, auto-rickshaws, cars, buses and freight vehicles with different footprints and queueing behaviour, weakening assumptions inherited from lane-disciplined car traffic. Crowdsourced travel-time and delay data offer broad spatial and temporal coverage, but do not directly identify whether a mismatch is caused by demand, capacity, route choice or intersection operation.

This study develops a MATSim-centred framework to test two calibration objectives within a common scenario: a traditional count-led model and a crowdsourced-first peak travel-time model (TTpeak). The framework converts MATSim event outputs into link- and mode-specific evidence, adjusts demand and network parameters between simulations, and retains the strongest calibrated state when subsequent adjustments deteriorate performance.

Meerut, Uttar Pradesh, provides the case study. The scenario contains 74 traffic analysis zones and seven motorised classes: motorcycles, auto-rickshaws, cars, buses, trucks, light commercial vehicles and other vehicles. Motorcycles represent 53.2% of the initial class share, cars 19.4%, auto-rickshaws 18.5%, and the remaining classes 8.9%. The road network was extracted from OpenStreetMap and refined to represent local topology and link attributes. Initial OD matrices and temporal profiles were derived using Rapidex, which estimates OD demand from pervasive travel-time data via a genetic-algorithm-based traffic assignment procedure (Waller et al., 2021). Classified video and manual counts provide volume targets and validation evidence, while crowdsourced observations provide link travel times and delays.

Before calibration, three empirical checks assessed whether the crowdsourced evidence represented plausible traffic processes. Delay-queue relationships produced R^2 values of 0.692-0.829 across signal groups. Speed-density analysis gave capacity estimates of 931-1,137 PCU/h/lane on unsignalised links and 889-1,689 PCU/h/lane on signal-controlled approaches. Temporal-profile correlations reached $r = 0.68$, with a median coefficient of variation of 9.7% across monitored approaches. However, crowdsourced volume indices were weak predictors of absolute flows and were therefore not used as the principal volume target.

2. MATSim Implementation and Calibration Design

The implementation uses MATSim 2026.0-2025w31. QSim ran with queue traffic dynamics and mode-specific vehicle types derived from vehicle data. Each fixed-mode population is assembled from separate OD matrices for the seven vehicle classes. The controller combines standard Charypar-Nagel activity, leg and stuck-agent scoring with a CaDyTS link-count correction, allowing observed counts to influence plan selection. SelectExpBeta preserves probabilistic plan choice, and ReRoute explores alternative paths during early iterations; rerouting is later disabled so each round can be scored against a more stable set of routes.

MATSim events feed custom listeners for aggregate and mode-specific volumes, travel times, and delays, as well as for stuck-agent and network diagnostics. The calibration controller evaluates successive MATSim iterations within each calibration round. It then selects the strongest-performing state, regenerates or adjusts mode-specific OD matrices, and updates bounded network levers: link capacity, lane availability and free-flow speed. Damping limits inter-round changes. If the composite score deteriorates materially, the workflow rolls back to the retained OD and network state, and the next adjustment step is reduced.

The traditional workflow is count-led. It evaluates each mode during the morning and evening peaks and seeks to include at least 85% of observations with GEH < 10. CaDyTS corrections remain part of the MATSim plan scoring, while the outer loop adjusts mode ODs and the shared network to account for count errors. Travel-time statistics are retained as a cross-metric diagnostic rather than the convergence objective.

The travel time-based model uses the same scenario and MATSim runner, but a different optimisation path. Its primary criterion is the proportion of mode-link observations for which the absolute difference between simulated

and observed travel time is no greater than the larger of 20% of the observed value or 10 s. An 85% pass rate is sought for every mode-peak combination. The iteration score combines continuous log travel-time error with a GEH penalty, so improved travel time cannot be selected without regard to implausible volumes. The TT-specific OD and network adjusters also use zero-flow, stuck-agent and feeder-bottleneck evidence. Thus, the two calibration approaches share a simulation infrastructure and bounded levers but use different logic for scoring, convergence, and adjustment.

In the evaluated configuration, motorcycle agents are assigned to the internal motorcycle_seepage mode, and QSim uses SeepageQ link dynamics. This represents smaller vehicles passing queued larger vehicles without strict first-in-first-out ordering (Agarwal et al., 2015; Agarwal and Laemmel, 2016). The alternative reversed-priority allocation is introduced below as a sensitivity experiment motivated by Meerut's motorcycle-dominated traffic composition.

Travel-time performance is reported using the modified TAG pass rate and the median absolute deviation from observed travel time. Volume performance is reported using GEH and percentage error. Convergence is assessed using the objective-specific mode-peak threshold, while aggregate AM and PM metrics permit a common comparison between workflows.

3. Results

Figures 1 and 2 compare simulated and observed conditions for the count-led and TTpeak calibration approaches. The volume panels present the morning and evening targets for the count-led model on separate axes. The travel-time panels show the TTpeak morning and evening observations together with the equality line and the modified TAG tolerance envelope.

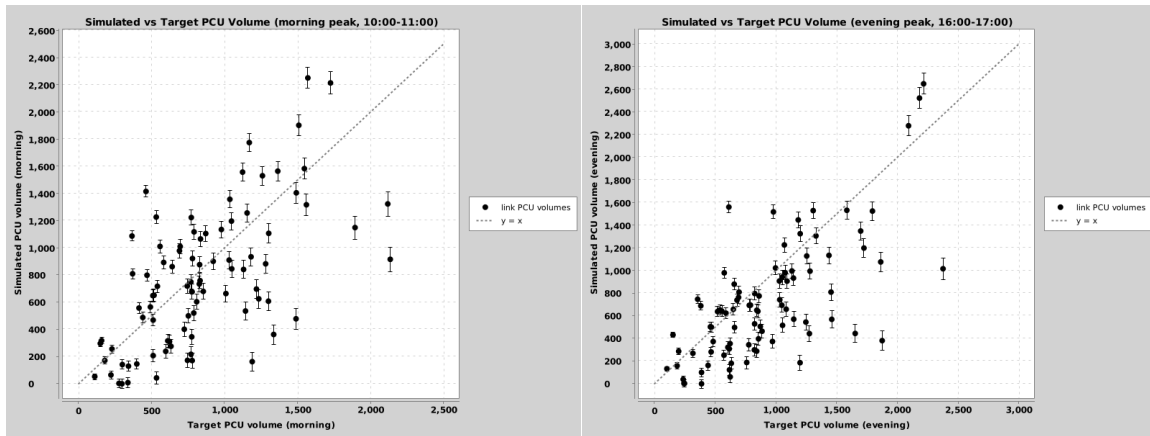


Figure 1. Simulated versus target PCU volumes for the traditional model: morning peak (left) and evening peak (right). Error bars show the 95% Poisson interval on the target flow.

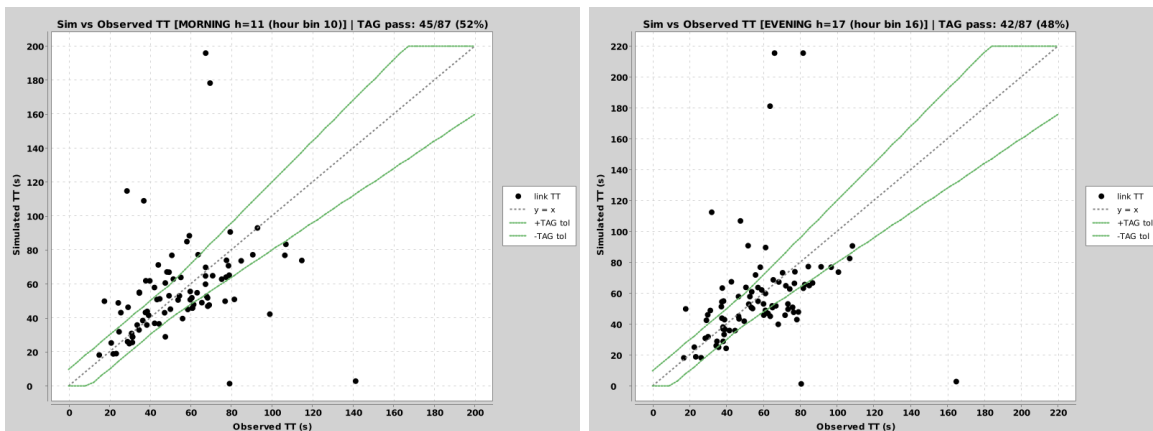


Figure 2. Simulated versus observed travel times for TTpeak: morning peak (left) and evening peak (right).

For the count-led model, 49.4% of the 176 peak link observations fall within $GEH < 10$, the median GEH is 10.2, and the coefficient of determination between simulated and target flows is 0.433. The evening peak is the stronger of the two, reaching 52.3% of observations within $GEH < 10$ and a median GEH of 9.3, with 31.8% of links within $GEH < 5$. For the travel-time model, 52% of morning and 48% of evening link observations satisfy the modified TAG criterion, and the median absolute deviation from observed travel time is 21.9% across both peaks, close to the 20% tolerance that defines the criterion.

The volume panels show the count-led model reproducing the overall flow level across the counted network, with the closest agreement in the mid-range of target volumes and the largest remaining differences concentrated on a small number of high-target links. The travel-time panels show a central cluster of links along the equality line and inside the modified TAG tolerance envelope, together with a smaller group of approaches whose simulated travel times sit above observation. These patterns identify where subsequent MATSim rounds can jointly address OD loading, feeder bottlenecks, route choice and local intersection representation.

4. Discussion and Conclusion

The study demonstrates how MATSim can support a controlled comparison of count-led and travel-time-led calibration in a heterogeneous, data-constrained setting. A common QSim and CaDyTS execution layer isolates the effects of changes to the outer objective, while event-based listeners and retained-state rollback make the calibration process auditable. Crowdsourced travel times broaden the evidence on congestion, but counts remain necessary to anchor absolute flows and diagnose demand, routing, and local intersection errors.

Meerut also raises a MATSim-specific question about seepage. Because motorcycles constitute the majority class, designating all motorcycles as the exceptional seepage stream can render seepage behaviour nearly universal. A reversed-priority experiment should therefore retain motorcycles as the principal queue stream and designate selected lower-share classes as seepage modes. This alternative is proposed for sensitivity testing and was not evaluated in the present comparison. Conventional and reversed allocations should be assessed against travel times, classified volumes, queues and stuck-agent patterns before selecting the better-calibrated representation.

Overall, the result plots demonstrate how the framework exposes different error structures under count-led and travel-time-led calibration. Future work will extend both calibration approaches and test whether alternative SeepageQ mode allocations improve the joint representation of travel times, volumes and queue formation in motorcycle-dominated traffic.

References

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